Selecting Reliable LLM Reasoning with a Neuro-Symbolic Trace Verifier

DASE7506 Advanced Machine Learning Methods - Mini-project 2

1. Research question and hypothesis

The main questions are: Do representations of full candidate reasoning help beyond answer agreement; should candidates be judged comparatively; and when does additional computation produce a reliable gain?

The hypothesis is that majority agreement is a strong signal, but candidates often share the same semantic mistake. A learned verifier using the supplied whole-trace embeddings should detect additional patterns associated with valid reasoning. A low-rank nonlinear projection should capture interactions missed by a linear model, while explicit equation checks should identify arithmetic failures.

Parameters are fitted only from training inputs and references. Validation is used for model selection. Each reported version was scored only after its configuration was frozen. V32 was selected from repeated training cross-validation and validation evidence without using any test outcome for training, tuning, or model selection.

2. Data and metric

The supplied bank contains 4,755 training, 798 validation, and 467 test questions. Each question has four candidate reasoning traces and a 1,024-dimensional representation for each trace. The output is exactly one candidate index per question. The official scorer parses the selected candidate's final numeric answer and awards a point when it equals the reference answer; it does not assess every reasoning step.

There are 379 validation questions on which all four parsed answers agree and 419 on which they disagree. Since every group contains at least one correct answer, all candidates in a unanimous group receive the same score. I report both complete validation accuracy and disagreement-only accuracy. The latter isolates questions where selection changes the outcome and is more comparable with the test bank, where every group contains at least two distinct valid parsed answers.

3. Method

3.1 Baselines

The first-position baseline always selects candidate 0. The answer-only baseline parses answers with the supplied numeric parser, groups equal fractions and decimals, and chooses the largest valid group. Ties go to the earliest candidate. This majority method uses no reasoning representation.

3.2 Surface-feature verifier

For each candidate I extract 23 small, auditable features: answer validity and group size, candidate position, text/word/line and number counts, coverage of numbers mentioned in the question, answer magnitude and integrality, final-answer markers, step/equation counts, and conclusion markers. This ablation tests how much can be gained from shallow formatting and lexical cues.

3.3 Relative embedding verifier

For candidate j of question i, I subtract the mean of that question's four supplied embeddings and L2-normalize the result:

$$r_{ij} = (e_{ij} - \text{mean}_k(e_{ik})) / ||e_{ij} - \text{mean}_k(e_{ik})||_2$$

This question-relative representation focuses the linear verifier on how a candidate differs from its alternatives. Each dimension is standardized with training statistics. A ridge model predicts whether a candidate's parsed answer equals the training reference:

$$\hat{w} = \text{argmin}_w ||Xw - y||_2^2 + \alpha ||w||_2^2$$

The embedding score is combined with an independently trained surface score:

$$b_j = \text{scale}(s_j^{(e)} + 0.25\, s_j^{(u)})$$

3.4 Nonlinear and arithmetic verifiers

A fixed 1,024 by 128 Gaussian projection generated with seed 1 supplies low-rank nonlinear interactions in the relative embedding:

$$z_{ij} = \tanh(0.5\, r_{ij} R)$$

A separate ridge model scores these features. The arithmetic verifier safely parses numeric equation chains written in each trace. A tolerant companion accepts common LaTeX and unit suffixes. A 31-feature block also describes answer rank, distance from the group median, sign, integrality, scale, and simple arithmetic reachability. Ridge and logistic models score these features. Candidate text is never executed. The scaled V6 component scores are fused as:

$$s_j = b_j + 0.475\, s_j^{(n)} + 0.1\, s_j^{(a1)} + 0.25\, s_j^{(a2)} + 0.5\, s_j^{(r)} + 0.9\, s_j^{(l)}$$

At inference, candidates with unparseable answers are masked. Candidates with the same parsed numeric answer form a group. Its score combines the strongest candidate evidence with a scale-stable corroboration term:

$$g(A) = \max_{j: a_j=A} s_j + 0.275 \log |\{j: a_j=A\}|$$

The best group is selected, followed by its highest-scoring candidate. The procedure accepts arbitrary bank sizes and never opens reference files.

V9 fits a 128-dimensional PCA basis to normalized question-relative trace embeddings; the retained axes explain 87.96% of training variance. It concatenates V6 group evidence with mean and maximum PCA trace features. A class-balanced RBF SVM with C=1 is trained on out-of-fold group features and contributes a 1.5-weight score. The PCA basis and 2,547 support vectors are evaluated with NumPy at inference.

3.5 Bounded symbolic program search

V32 extracts up to eight quantities from the question and constructs arithmetic programs with addition, subtraction, multiplication, and division. Programs use at most five input occurrences, while a target-aware beam retains at most 60 values per number subset. This bounds runtime without using question IDs or references.

Nineteen candidate features summarize exact reachability, direct occurrence, program size, input coverage, operation types, reachable-program counts, closest-value distance, and integrality. Mean and maximum features are pooled within equal-answer groups and combined with compact semantic and executed-trace evidence by a 12-unit tanh listwise ranker:

$$s_j^{(V32)} = s_j^{(V9)} + 1.5\, s_j^{(program)}$$

The search does not reproduce a natural-language rationale. It tests the narrower and more robust condition that an answer can be constructed by a short arithmetic program over quantities explicitly present in the problem.

3.6 Selection protocol

I compare surface features, embeddings, their concatenation, candidate-level selection, answer-group aggregation, direct listwise training, late fusion, random nonlinear features, arithmetic features, numeric structure, alternative targets, ensembles, nearest-neighbour and lexical models, stacking, wider kernels, pairwise ranking, executed-trace provenance, consensus selectors, reference-plan prediction, and train-only easy-positive augmentation. New components are admitted only when validation and repeated training-only comparisons support them. V32 produced the strongest full validation result and improved V9 on informative questions under all three repeated partitions. Its configuration was frozen before public-test scoring.

4. Results

Table 1. Validation accuracy and disagreement-only accuracy.

Method	Val. correct	Val. acc.	Disagree correct	Disagree acc.
First candidate	624/798	78.20%	245/419	58.47%
Answer majority	688/798	86.22%	309/419	73.75%
Surface features	706/798	88.47%	327/419	78.04%
Candidate embedding	727/798	91.10%	348/419	83.05%
Answer-group embedding	732/798	91.73%	353/419	84.25%
Surface + embedding	726/798	90.98%	347/419	82.82%
Late-fused answer group (V3)	736/798	92.23%	357/419	85.20%

Method	Val. correct	Val. acc.	Disagree correct	Disagree acc.
Nonlinear embedding fusion	740/798	92.73%	361/419	86.16%
Nonlinear + arithmetic (V4)	742/798	92.98%	363/419	86.63%
Structured ridge (V5)	746/798	93.48%	367/419	87.59%
Ridge + logistic numeric (V6)	746/798	93.48%	367/419	87.59%
Random-projection RBF (V8)	747/798	93.61%	368/419	87.83%
PCA semantic RBF (V9)	748/798	93.73%	369/419	88.07%
Program search + listwise (V32)	**752/798**	**94.24%**	**373/419**	**89.02%**

V32 improves over majority by 64 validation questions, or 8.02 percentage points; on disagreement questions the gain is 15.27 points. It improves V9 by four validation questions, all from the disagreement subset. The surface-only model trails the embedding verifier by 2.63 points, so formatting cues do not account for the main gain.

The V32 program-search ranker improves V9 on informative train/CV questions by +8, +7, and +5 under the default, salt-A, and salt-B partitions; corresponding informative validation gains are +4, +6, and +5. After freezing the configuration, the official public-test result is 392/467 = 83.940043%. Test examples and per-example outcomes are not used for training or parameter selection.

4.1 Controlled ablation study

Table 2 reports matched comparisons that isolate key design choices. Positive deltas mean that restoring the named component improves over its ablated method; the rows are controlled pairs rather than an undifferentiated version history. Every pair uses the same frozen validation split and scorer. Because all 379 unanimous-answer groups are correct under any selection, full-set count changes equal disagreement-set count changes; the latter is the more sensitive metric.

Table 2. Controlled component ablations on validation data.

Component restored	Ablated	With component	Val. delta	Disagree delta
Full-trace representation	Surface: 706	Embedding: 727	+21 (+2.63 pp)	+21 (+5.01 pp)
Answer-group comparison	Candidate: 727	Grouped: 732	+5 (+0.63 pp)	+5 (+1.19 pp)
Arithmetic checking	Nonlinear: 740	V4: 742	+2 (+0.25 pp)	+2 (+0.48 pp)
Logistic calibration	V5: 746	V6: 746	0 (0.00 pp)	0 (0.00 pp)
Bounded symbolic module	V9: 748	V32: 752	+4 (+0.50 pp)	+4 (+0.95 pp)

Reasoning representation versus surface cues. Replacing the 23 shallow features with the supplied full-trace representation adds 21 correct choices. On disagreement questions, errors fall from 92 to 71, a 22.8% relative reduction. Length, formatting, equation counts, and conclusion markers therefore do not explain most of the gain. This does not prove that every embedding dimension encodes valid reasoning: latent stylistic cues may remain. It establishes the narrower claim that the whole-trace representation contains information beyond the audited surface block.

Comparative judgement. Answer grouping raises the result from 727 to 732 correct. All five recovered decisions are disagreements, reducing their errors from 71 to 66 (7.0%). This supports the hypothesis that alternatives provide useful context. Grouping cannot help unanimous groups and may reinforce a shared mistake, so the method treats agreement as corroboration rather than a correctness rule.

Arithmetic evidence and model capacity. The strict arithmetic checker adds two correct decisions and reduces disagreement errors from 58 to 56 (3.4%). Its coverage is limited to equations explicitly present in a trace and cannot expose a misread condition followed by consistent arithmetic. Logistic numeric calibration in V6 changes no top-1 decision. Its score may still inform later fusion, but the ablation gives no evidence for an independent accuracy gain and shows that added nonlinear capacity need not help.

Bounded symbolic module. V9 is the ablation that removes program-search features and their 12-unit listwise scorer. V32 changes 11 selected answers: seven repair an error, three introduce an error, and one leaves correctness unchanged, for a net gain of four. Disagreement errors fall from 50 to 46, an 8.0% relative reduction. The regressions show that reachability is evidence, not proof.

The single-split gain is small, so repeated evidence matters. Under the default, salt-A, and salt-B partitions, V32 improves informative train/CV questions over V9 by +8, +7, and +5; corresponding informative-validation gains are +4, +6, and +5. The consistent direction is stronger than the +4 headline alone, although repeated use of the same validation bank still leaves some model-selection bias.

Accuracy-cost trade-off and scope. Fresh program-feature search takes 257.07 seconds, 84.0% of the 306.11-second clean reproduction process. The frozen V32 asset is 4.78 MiB and complete local prediction plus scoring takes 18.23 seconds. The accuracy return is modest for most of the reproduction-time computation, but the module is retained because gains repeat and deployment stays within the 16 MiB and 30-second limits. This ablation removes search and learned fusion together; it does not attribute their contributions separately.

5. Resource use and reproducibility

Training uses fixed seeds 1 (random projection and PCA) and 8200 (listwise ranker). On an AMD Ryzen 9 8945HX, a clean base train, V9 export, fresh V32 search/cache build, and V32 export take 17.84, 13.20, 257.07, and 19.00 seconds respectively (306.11 seconds total). The regenerated model produces identical validation and test choices. With OMP, OpenBLAS, and MKL thread counts fixed to four, fresh V32 prediction followed by independent scoring takes 18.23 seconds. The reference Xeon remains authoritative.

The only inference asset is model.npz (5,013,724 bytes, 4.78 MiB). Its SHA-256 is `b384830a79de9d98e204d4273987b0ea7afaf0e84c008bdb0758a06d610af870`. It stores the PCA basis, compact verifiers, and their training statistics. The method uses only NumPy at inference and performs no network access, generation, retraining, ID lookup, or assumption about bank size.

6. Limitations and error analysis

The supplied embedding summarizes the whole trace, so the experiment shows that the representation is useful but cannot identify which individual reasoning step caused a score. The surface ablation reduces one alternative explanation, yet unmeasured stylistic cues may remain embedded. A stronger process-level study would need step-level representations or a permitted way to recompute embeddings after controlled trace masking; neither is supplied here.

The arithmetic checker validates equations that candidates explicitly write, but a trace can remain self-consistent while misreading the problem or omitting a condition. The bounded search is intentionally incomplete: it uses the first eight quantities, at most five inputs, four elementary operations, rounded states, and a finite beam. It does not directly model units, implicit constants, language-only rates, or longer programs. Test accuracy remains below validation disagreement accuracy, indicating distribution shift. Future work should add unit- and relation-aware symbolic states.

7. Conclusion

A 4.78 MiB verifier combining relative full-trace embeddings, PCA semantic routing, arithmetic checks, numeric structure, parsed-answer grouping, and bounded symbolic search substantially outperforms fixed-position and majority baselines. The final V32 selector reaches 94.24% validation and 83.94% frozen public-test accuracy within the CPU and asset limits.

AI assistance and reuse disclosure. OpenAI Codex provided substantive assistance in interpreting the assignment, implementing and testing the method, running the documented experiments, and drafting this report. The submitter reviewed the resulting code and remains responsible for understanding and explaining it. Frozen inference uses NumPy, while training additionally uses scikit-learn. No external dataset, pretrained weight, third-party solution, or additional model asset is included.